

Gynecology

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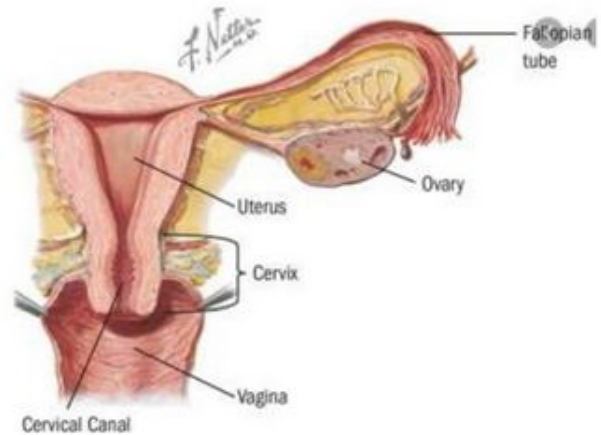
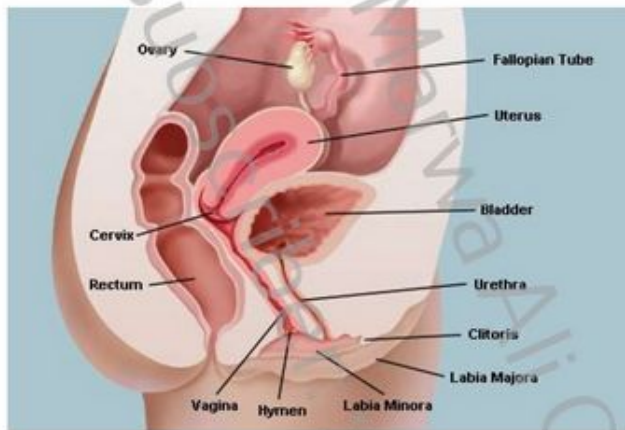
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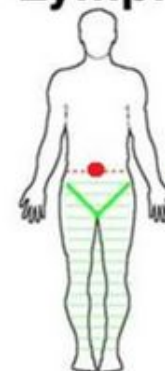
1- Anatomy of the female genital system



➤ Lymphatic drainage:

- *Scrotum , labia and external genitalia >>> inguinal LNs
- *Glans and clitoris >>> external iliac LNs
- *Testis and ovaries >>> para-aortic LNs

Lymph



1. Superficial Inguinal

everything below the umbilicus

2 exceptions!

- gonads
- para aortic
- lateral foot/
- popliteal >> inguinal

Examoscope:

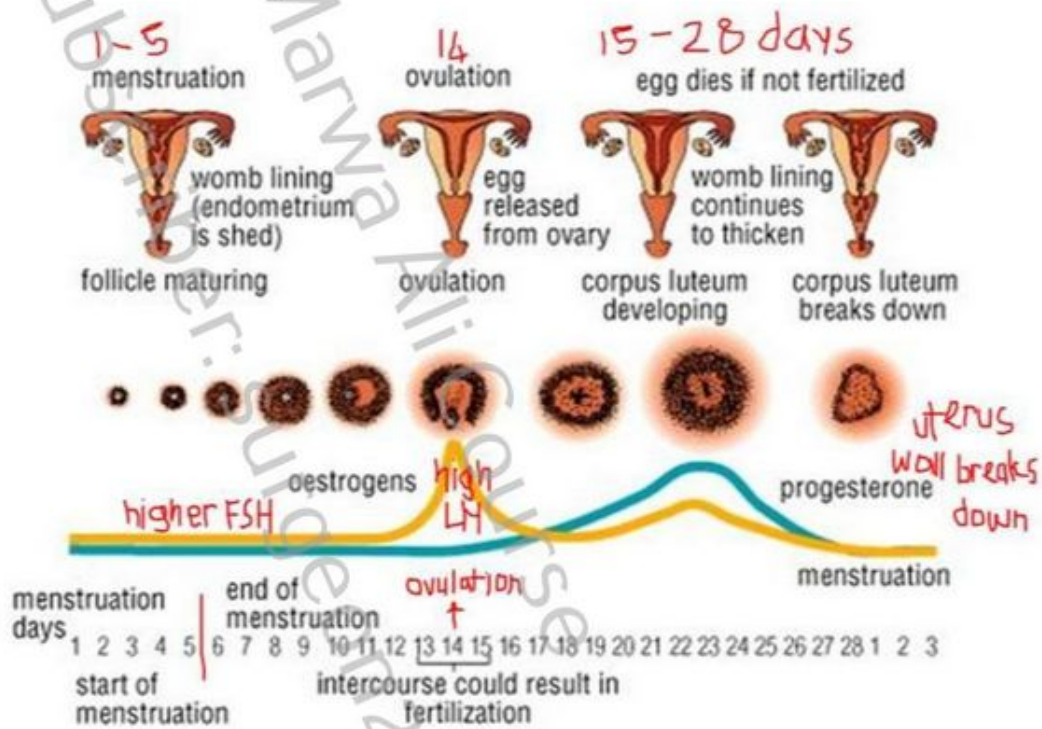
A 56 year old lady was diagnosed with a right sided ovarian cancer.
LYMPH NODES MOST LIKELY to be involved?



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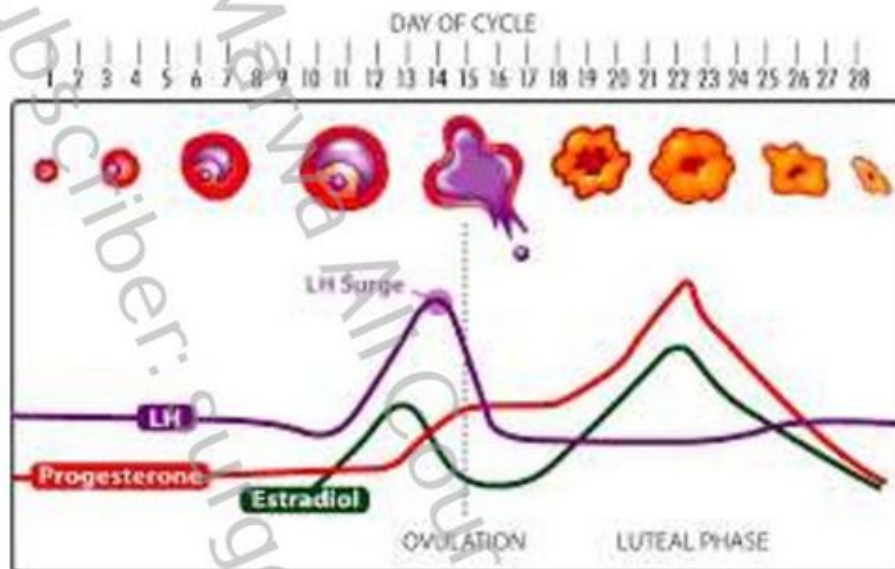
2- Hormonal cycle:



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3- Testing ovulation (day 21 progesterone):



- ✓ Serum progesterone (Mid-luteal cycle progesterone) is tested one week before a menstrual period is expected. Example, on day 21 of a 28 day menstrual cycle. (28 days - 7 days = day 21).
- ✓ A serum progesterone more than 30 nmol/L is consistent with ovulation and no further biochemical assessment is required.

Don't forget day 21 in a 28 days period >>>>

If the period is a number rather than 28, just subtract 7 from it (1 week before menstruation)

e.g 31 days period >>> 31-7= 24 >>> day 24 progesterone level.

If the level is low >>> repeat the test (ovulation does not happen every month)



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As part of the management for infertility, FSH and LH are usually taken in clinics >>> measured if there is menstrual irregularity >> to exclude conditions like premature ovarian insufficiency (LH and FSH both raised), polycystic ovarian syndrome (LH:FSH ratio increased) and hypogonadotropic hypogonadism (LH and FSH reduced).

Assessment of infertility:

- a- Mid luteal progesterone
- b- LH & FSH
- c- Hysterosalpingiography

Causes of infertility:

- a- Ovarian failure
- b- Uterus (fibroid, adhesions)
- c- Tubal damage (e.g PID complications)
- d- Unexplained

Examoscope:

A lady presents with difficulty in conceiving +for more than 2. There is no previous history of PID. She has a BMI of 23. She has a regular 32 day menstrual cycle. What is the SINGLE most appropriate test to perform to assess ovulation?



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4- Amenorrhea:

- ✓ **primary** (failure to start menses by the age of 16 years or 14 in absence of 2ry sexual ccc)
- ✓ **secondary** (cessation of established, menstruation for 6 months or longer).

Causes of amenorrhea:

- **Pregnancy**
- **Post pill amenorrhea**
- **congenital malformations (Mullerian agenesis / imperforate hymen)**
- **polycystic ovarian syndrome (PCOS)**
- **premature ovarian failure**
- **Sheehan's syndrome**
- **Asherman's syndrome (intrauterine adhesions)**
- Turner's syndrome
- hypothalamic amenorrhea (e.g. Stress, excessive exercise)
- congenital adrenal hyperplasia
- hyperprolactinaemia
- thyrotoxicosis or hypothyroidism

Initial investigations:

- exclude pregnancy with urinary or serum bHCG
- gonadotrophins: ↓ levels indicate a hypothalamic cause / ↑ levels suggest an ovarian problem (e.g. Premature ovarian failure)
- prolactin
- androgen levels: raised levels may be seen in PCOS
- oestradiol
- thyroid function tests
- vaginal examination and pelvic US to exclude anatomical causes



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INVESTIGATING PRIMARY AMENORRHEA

SITE OF DISORDER	DIAGNOSIS	INVESTIGATIONS
HYPOTHALAMUS	Hypothalamic-hypogonadism	FSH, LH and estradiol - Low
PITUITARY	Pituitary adenoma	Prolactin - High FSH, LH and estradiol - Low
OVARY	Gonadal dysgenesis (Turner's syndrome)	FSH and LH - High Estradiol - Low Karyotype - 45 XO
MULLERIAN TRACT	Absent uterus (Testicular feminization)	FSH, LH, estradiol - normal
GENITAL TRACT	Imperforate hymen	FSH, LH, estradiol - normal PCT - negative Examination - imperforate hymen

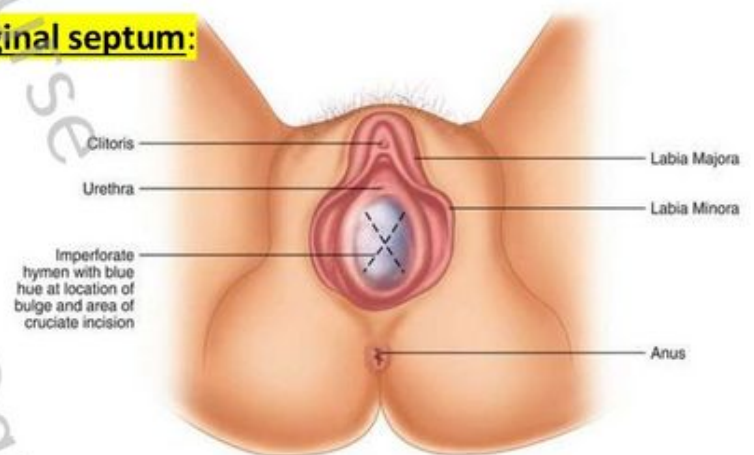
Imperforate hymen and transvaginal septum:

Primary amenorrhoea

Presence of 2ry sexual CCC

Cyclic pain+ lower abd mass

>>> **hematometra** (accumulation
Of blood within the uterus)
a **bluish bulging** membrane



examoscope:

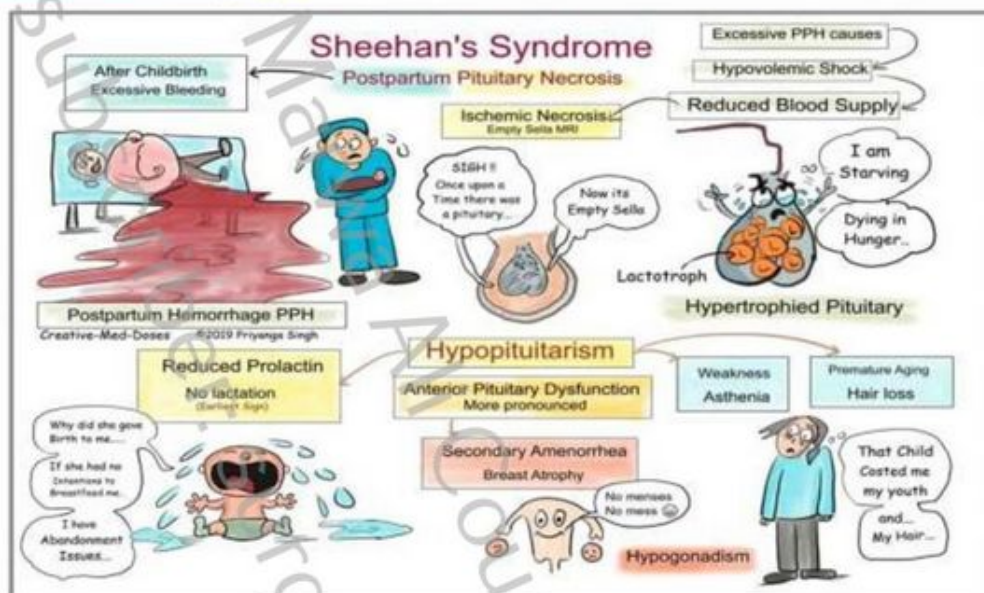
a femlae 16 yrs/never had her menses/ presence of 2ry sexual ccc/
cyclic abdominal pain/ examination: bluish bulge >> diagnosis??



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Sheehan's syndrome:



- ✓ **Obstetric hge** >>> pituitary necrosis >>> panhypopituitarism
- ✓ The 1st sign is >>> **failure of lactation** (↓ prolactin)
- ✓ ↓ cortisol >>> addison crisis

Examoscope: a lady after giving birth, suffered from postpartum he + needed blood transfusion/ suffer from amenorrhae/ can't produce any breast milk

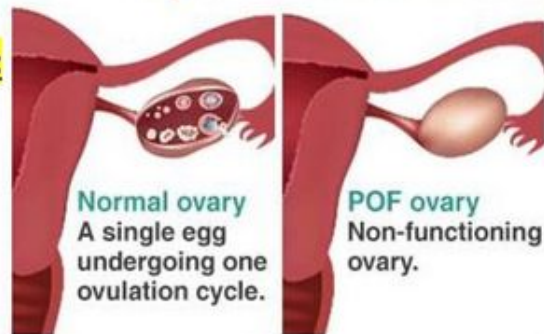


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One of the hidden causes of early menopause is **premature ovarian failure (POF)**, which refers to ovaries problems before the age of 40.

Premature ovarian failure:



Premature ovarian failure is defined as the **onset of menopausal symptoms and elevated gonadotrophin levels before the age of 40 years.**

Causes:

- **idiopathic - the most common cause**
- **chemotherapy, radiation** (exam scenario)
- autoimmune
- bilateral oophorectomy (surgical menopause)

Features >> similar to those of menopause

- hot flushes, night sweats, vaginal dryness and dyspareunia
- infertility
- secondary amenorrhoea and oligomenorrhoea
- **raised FSH, LH levels FSH >25 IU/L in 2 occasions, 4 weeks apart) LH high, Estrogen <50 Low, Prolactin normal**

TTT : HRT until age 51

Examoscope:

* A 36yrs lady presented with amenorrhoea, hot flushes, vaginal dryness
>>> appropriate test???

** A 35 yrs lady came to the infertility clinic, her lab >> FSH was 50
>>>diagnosis ???



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5- PCO (Rotterdam criteria) 2 of 3:

Features:

- subfertility and infertility
- **menstrual disturbances:**
(oligomenorrhea and amenorrhoea)
- **hirsutism, acne** (due to hyperandrogenism)
- **obesity**
- acanthosis nigricans: due to insulin resistance

Investigations:

- pelvic ultrasound: **multiple cysts** on the ovaries
- FSH, LH, prolactin, TSH, and testosterone are useful investigations: **raised LH:FSH ratio** is a 'classical' feature. Prolactin may be normal or mildly elevated. Testosterone may be normal or mildly elevated
- check for impaired glucose tolerance

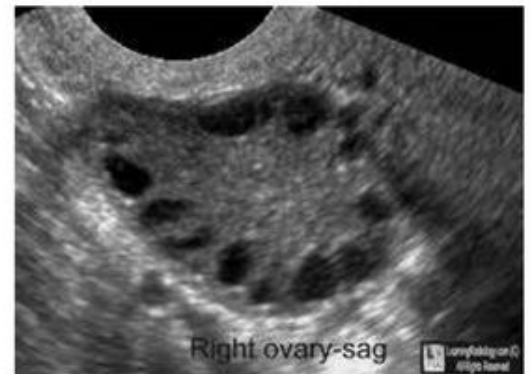
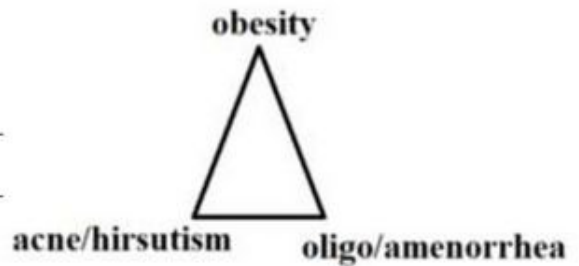
Diagnosis : (Rotterdam consensus criteria Two out of three)

1. **Ultrasound** >>> Polycystic ovaries (either 12 or more follicles or increased ovarian volume)
2. **Oligo-ovulation or anovulation**
3. **Clinical and/or biochemical** signs of hyperandrogenism

Management:

General:

- **weight reduction** if appropriate
- if a women requires contraception then (COC) pill may help regulate her cycle or IUS



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Infertility:

- **weight reduction** if appropriate
- **clomifene citrate** (1st line)
- $\pm$ metformin ??
- **Surgical : laparoscopic drilling**

Management:

Wt loss wt loss wt loss

Cocp, IUS

Clomifene citrate

Laparscopic drilling

Metformin??

PCOS SYMPTOMS:



POLYCYSTIC OVARY SYNDROMES (PCOS)

Is it really PCOS?

PCOS	Hypothalamic Amenorrhea
absent or irregular cycles	absent or irregular cycles
polycystic ovaries	polycystic ovaries
elevated androgens	low-normal hormone levels
can have insulin resistance	history of under eating or low carb dieting
elevated LH	low/normal LH

Examoscope:

** Young female/ infertility/ high BMI/ acne/ hirsutism/ hyperpigmentation of axilla skin fold >>> diagnosis??

** Same scenario, want to conceive >> 1st line management?? 1st line medical management???



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6- Endometriosis:

Clinical features >>> D features:

- Deep chronic pelvic pain
- Dysmenorrhoea
(pain often starts 1-2 days before bleeding)
- Deep dyspareunia
- subfertility (Decreased fertility)
- non-gynaecological:
 - Dysuria, urgency, haematuria.
 - Dyschezia (painful bowel movements)

Investigation:

- *** laparoscopy is the gold-standard investigation (not hysteroscopy)
- ** US to exclude ovarian endometrioma (do not exclude)
- * MRI, preoperative to exclude deep endometriosis

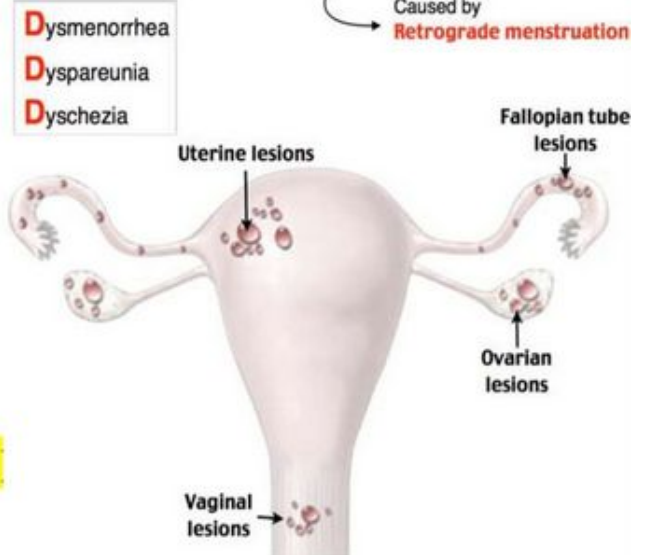
Management :depends on clinical features :

- ✓ symptomatic relief >>> NSAIDs and/or paracetamol (recommended 1st -line treatments)
 - ✓ combined oral contraceptive pill or progestogens e.g. medroxyprogesterone acetate should be tried (treat suspected cases empirically with CoCp and progesterone for 3-6 months before laparoscopy (as long as fertility is not an issue))
 - ✓ If analgesia/hormonal treatment does not improve symptoms or if fertility is a priority >>> refer to 2nd care.
- >>> Secondary treatments include: laparoscopic excision and laser ttt.

Endometriosis

Abnormal growth of endometrium outside of the uterus

Caused by
Retrograde menstruation



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Symptoms of endometriosis may overlap with symptoms of PID >>> but in PID NO dyschezia + general symptoms of infection may be mentioned like cervical excitation, fever, elevated WBCs,....

N.B: Myometrial endometriosis = adenomyosis

Exam scope:

A young female c/o with painful micturation, dyspareunia/ painful voiding of her bowels >>> diagnosis?? Gold standard investigation?? Management??



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7- Ovarian cancer CA 125:

Risk factors

- family history: mutations of the BRCA1 or the BRCA2 gene
- a 1st degree or 2nd degree relative >> carries a gene mutation
- 2 family members who are 1st degree relatives of each other have ovarian cancer
- 1 family member has both breast and ovarian cancer

SIGN guidelines advise that women with a family history that appears to increase their risk of developing ovarian cancer should be referred to a clinical genetics service for full assessment of risk.

- many ovulations: early menarche, late menopause, nulliparity
- Smoking & age

Clinical features are vague:

- abdominal distension and bloating
- abdominal and pelvic pain
- urinary symptoms e.g. Urgency
- early satiety
- diarrhoea

SYMPTOMS OF OVARIAN CANCER



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Steps to investigate a patient with suspected ovarian cancer:

1- CA125 (initial)

- if the CA125 is raised (≥ 35 IU/mL) >> urgent ultrasound scan of the abdomen and pelvis should be ordered
- a CA125 should not be used for screening for ovarian cancer in asymptomatic women

2- pelvic and abdominal ultrasound scans

3- CT scan of the pelvis and abdomen

4- MRI scan can be used for pre-operative staging

5- High-risk women may be offered a referral for prophylactic salpingo-oophorectomy

>>>>> If Diagnosis is difficult >>> diagnostic laparotomy

Condition	Tumor marker
Seminoma	LDH
Teratoma	Alpha fetoprotein
Choriocarcinoma	B HCG
Prostate cancer	PSA
Ovarian cancer	CA 125

Management:

usually a combination of surgery + chemotherapy

examscope:

60 yrs lady with bloating & vague abd pain/ exam: adnexal mass >>> investigation??

55yrs lady, her mother died with ovarian cancer, she is smoker and on HRT, what is the most important risk factor??



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Meigs' syndrome (Demons-Meigs syndrome)

- A benign ovarian tumour (pelvic pain)
- Ascites (abdominal distention)
- Pleural effusion (shortness of breath)



Meigs' syndrome is rare and is a diagnosis of exclusion.

The symptoms resolve after removal of the ovarian mass.

The effusion is exudative

Light's Criteria: Pleural fluid is considered to be an exudate if any of the following are true:

- Pleural fluid/serum protein > 0.5
- Pleural fluid/serum LDH > 0.6
- Pleural fluid LDH $>$ two-thirds of the upper limits of normal serum value

a pleural effusion is considered a transudate if the fluid protein is < 30 g/L and an exudate if the fluid protein is > 30 g/L

examoscope:

A lady presented with dyspnea, pelvic pain, on exam: percussion: dullness over the chest, shifting dullness in her abdomen >> diagnosis??

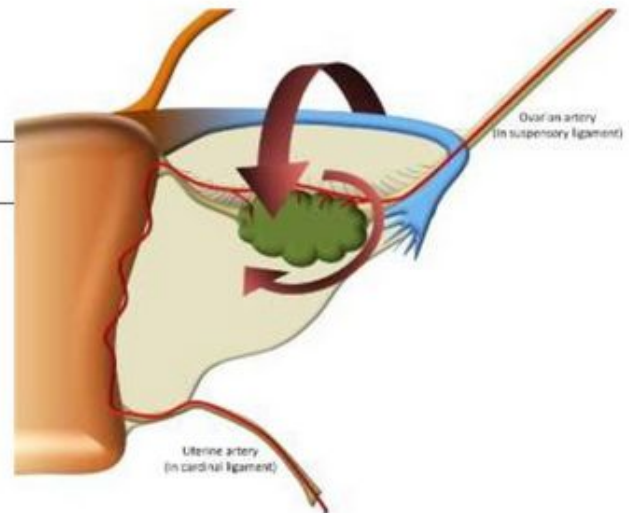


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8- ovarian torsion

Ovarian torsion is a gynaecological emergency and required urgent surgical management to prevent further necrosis of the ovary.



Any large adnexal mass puts the patient at risk of ovarian torsion.

Features:

- ❖ Sudden onset unilateral lower abdominal pain. Onset may coincide with exercise.
- ❖ Nausea and vomiting are common
- ❖ Unilateral, tender adnexal mass on examination
- ❖ A raised white cell count and fever can be seen in ovarian torsion when the ovaries start to become necrotic.

The ultrasound finding: **large cyst and reduced vascularity**

Even if the ultrasound scan reveals normal vascularity >>> don't rule out ovarian torsion and the management would remain the same.why??

- Intermittent torsions may have normal vascularity at times.
- There may be dual blood supply from both the ovarian and uterine arteries.

If suspicion is high, the patient needs to go for surgery under the gynae team.



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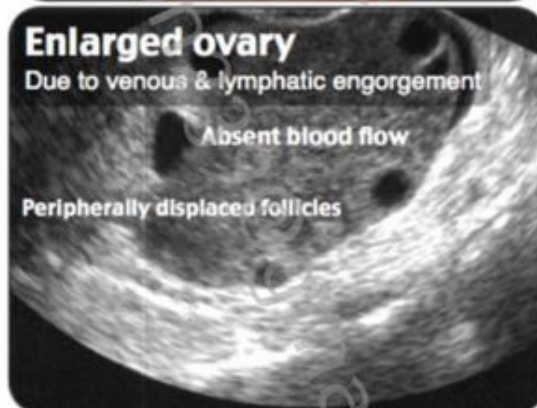
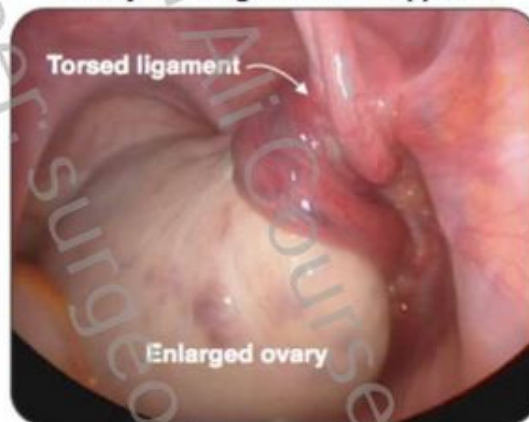
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Examoscope:

a female presented with sudden onset severe iliac fossa pain with nausea and vomiting, her pregnancy test is – ve and US exam revealed diminished ovarian blood flow?? Diagnosis?? Next step??

Ovarian Torsion

→ Complete or partial rotation of the ovary on its ligamentous supports



Risk factor

- Ovarian mass ≥ 5 cm
- Reproductive age
- Pregnancy
- Ovulation induction
- Prior torsion

Clinical

- Pelvic pain
- Adnexal mass
- Nausea/vomiting
- Fever
- Vaginal bleeding

Imaging

- Pelvic ultrasound

Management

- Operative evaluation



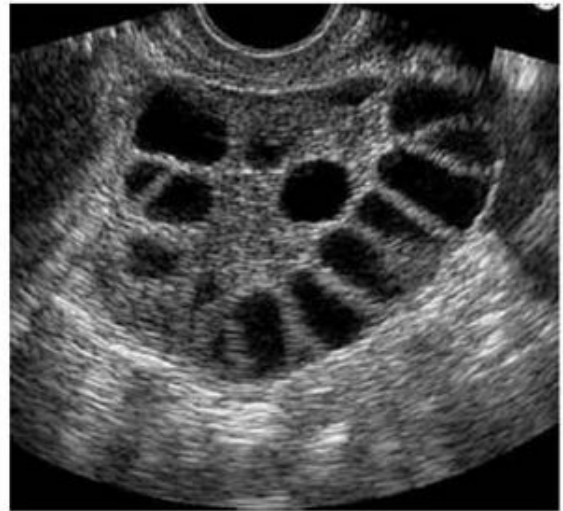
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9- Ovarian lesions by US:

1- PCO (cookies):

- Small follicles arranged at the periphery
- Central white stroma
- Large ovarian volume



2- Endometriosis (Chocolate cyst)

- Thick wall unilocular cyst
- ground glass appearance



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3- Dermoid cyst (iceberg)

- ❖ Iceberg tip sign
- ❖ Fat fluid level
- ❖ Unilocular
- ❖ Dermoid mesh



4- Ovarian teratoma

echogenic tubercle projecting into
cystic lumen



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5- Tuboovarian abscess

- ❖ Multilocular
- ❖ Separations
- ❖ Thick wall
- ❖ Echogenic debris



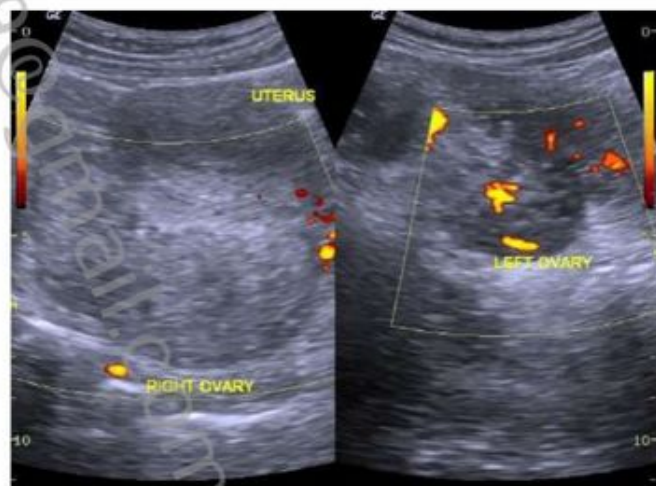
6- Theca lutein cyst:

Bilateral cystic masses



7- Ovarian torsion:

Diminished blood supply on
color Doppler



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10- Vaginal infections: (commonly asked topic)

The main complaint is **VAGINAL DISCHARGE**

a- Bacterial Vaginosis:

- ✓ caused by overgrowth of mixed anaerobes, such as **Gardnerella vaginalis** >>> replaces the dominant vaginal lactobacilli >>> result in a raised vaginal pH >>> allows production of proteolytic enzymes >>> degrade peptides into offensive-smelling products and increases discharge.
- ✓ Although Bacterial Vaginosis "Gardnerella Vaginalis" is **not an STDs**, it is **the most common cause of abnormal vaginal discharge** in childbearing age.
- ✓ Thin, grey-white, fishy (**VERY offensive**) smelling discharge.
- ✓ Vaginal **itching is uncommon**.
- ✓ Rx -> Metronidazole + Clindamycin

✓ Amsel's Criteria :

3 of 4 criteria are diagnostic for Bacterial Vaginosis

- 1) Homogenous grey-white **discharge**.
- 2) When adding Potassium Hydroxide 10% (KOH) to the discharge >>>> fishy smell (**Whiff test**).
- 3) "**Clue Cells**" under microscopy.
- 4) Vaginal **pH** > 4.5

b- Trichomonas vaginalis:

- ✓ Frothy, yellowish-greenish smelly vaginal discharge.
- ✓ Vaginal **itching** is common.
- ✓ **Strawberry Cervix**.
- ✓ Vaginal pH: > 4.5
- ✓ Rx -> Metronidazole



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c- **Candida vaginitis:**

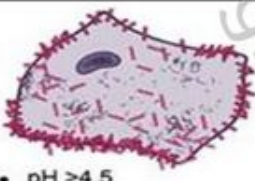
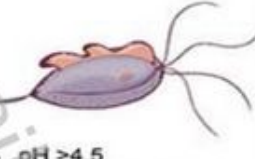
Thick white discharge + itching

RFs: * Pregnant women

* recent use of antibiotics (broad spectrum antibiotics

>>> destroy the normal flora leading to the development of vaginal thrush.

N.B: broad-spectrum antibiotics inhibit lactobacilli and disrupt the vagina's normal flora which can predispose women to develop either vaginal candidiasis or bacterial vaginosis due to the change of the natural balance of bacteria in the vagina.

Differential diagnosis of vaginitis			
Diagnosis	Bacterial vaginosis (<i>Gardnerella vaginalis</i>)	Trichomoniasis (<i>Trichomonas vaginalis</i>)	Candida vaginitis (<i>Candida albicans</i>)
Examination	 - Thin, off-white discharge with fishy odor - No inflammation	 - Thin, yellow-green, malodorous, frothy discharge - Vaginal inflammation	 - Thick, "cottage cheese" discharge - Vaginal inflammation
Laboratory findings	 - pH >4.5 - Clue cells - Positive whiff test (amine odor with KOH)	 - pH >4.5 - Motile trichomonads	 Normal pH (4-4.5)
Treatment	Metronidazole clindamycin	Metronidazole; treat sexual partner	local clotrimazole

KOH = potassium hydroxide.



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Understanding the gynae swabs: (for reading):

Types of swabs:

vulvovaginal swab	endocervical swab	high vaginal swab
either self collected (the most sensitive because allow adequate sampling), or doctor collected		
screening for chlamydia and gonococcus by NAAT	culture for gonococcus in stuart media	detect pathogens like BV, TV, Candida and group B strept

What to choose ????

✓ If the patient is presented to sexual health clinic:

- 1- The doctor will perform a high vaginal swab, use his microscopy, put the sample on slides (dry mount and wet mount) to detect pathogens like BV, TV, candida >> directly treat
- 2- Send another Vulvovaginal swab for NAAT >>>> if the NAAT is positive for chlamydia >>> treat
If the NAAT is +ve for Gonococcus >>>>
- 3- Take a high vaginal and an endocervical swab for culture in a stuart media (as gonocci have resistant strains)

✓ If the patient is presented in a GP clinic

>>>> Vulvovaginal swab and send for NAAT



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11- Cervicitis:

❖ infection, irritation or inflammation of the cervix and most likely seen in young woman with new sexual partners.

- ✓ Usually asymptomatic
- ✓ Can present with vaginal discharge, lower abdominal pain
- ✓ intermenstrual bleeding or post coital bleeding

- **Diagnosis:** Vulvovaginal swab with NAAT >>>> If NAAT results show +ve for gonorrhoea >> a culture needs to be sent prior to starting antibiotics if possible (test for susceptibility & resistant strains).

Endocervical swab and a high vaginal swab in transport medium (charcoal preferably) is sent for gonorrhoea culture if the patient has a positive NAAT

- **Screening (asymptomatic):** self collected Vulvovaginal swab with NAAT
- **screening asymptomatic heterosexual males for gonorrhoea:** NAAT >> a first pass urine is collected for males

management

Chlamydia cervicitis: DA

- **first line** >>> Doxycycline 100mg 2X7
- **second line** >>> Azithromycin 1g orally as a single dose, followed by 500mg once daily for two days
- **Pregnant** >>> erythromycin

Neisseria gonorrhoeae cervicitis: C or C

- When **antimicrobial susceptibility not known** prior to treatment >>> Ceftriaxone 1g IM single dose
- If **antimicrobial susceptibility** to ciprofloxacin >>> Ciprofloxacin 500mg PO single dose



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Chlamydia in men >>> urethritis

The major complication of untreated chlamydia "and N. Gonorrhea" in males is: (Epididymo-Orchitis) or (Epididymitis).

The major complication of untreated chlamydia "and N. Gonorrhea" in Females is: (Salpingitis).

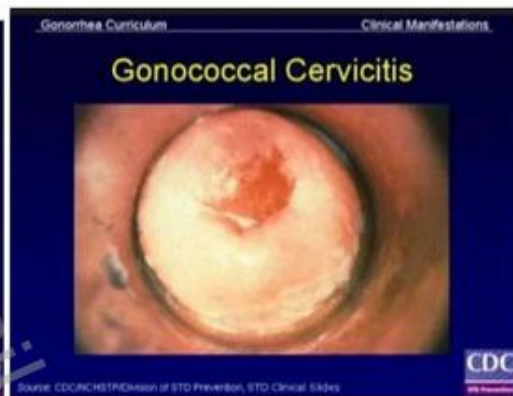
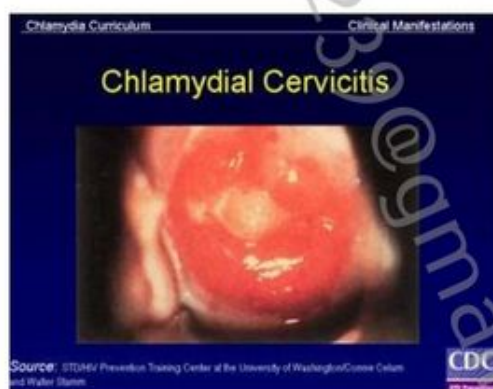
women with vaginal discharge at risk of a sexually transmitted infection: suspected organisms???

A vulvovaginal swab AND a high vaginal swab is routinely done in sexual health clinics for.

***Vulvo-vaginal swabs >>> for detecting Neisseria gonorrhoea and Chlamydia trachomatis.**

***A high vaginal swab from the posterior fornix >>> to detect Trichomonas vaginalis.**

Trichomonas vaginalis is diagnosed by observing the motile organism on microscopy. A high vaginal swab could also look for other organisms like BV and yeast infections.

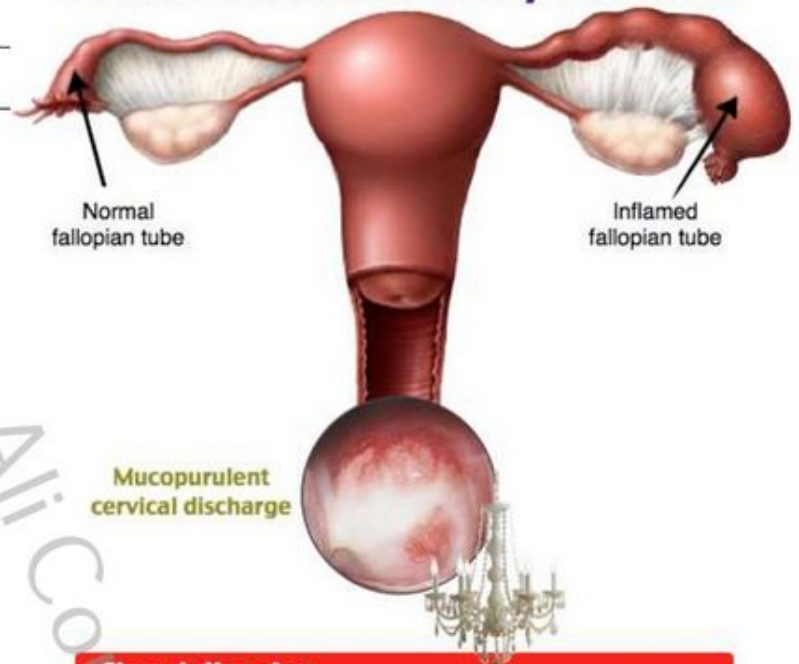


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Pelvic Inflammatory Disease

12- PID



Chandeller sign

Severe cervical motion tenderness seen in PID

Major Criteria

- Lower abdominal pain
- Lower abdominal tenderness
- Cervical motion tenderness
- Adnexal tenderness

Supporting Criteria

- Fever $> 38^{\circ}\text{C}$
- Abnormal vaginal discharge
- Positive culture for *N. gonorrhoeae* or *C. trachomatis*
- WBC $> 10,000$
- Elevated CRP or ESR

Pelvic inflammatory disease (PID): infection & inflammation of the female pelvic organs including the uterus, fallopian tubes, ovaries & the surrounding peritoneum. It is usually the result of ascending infection from the endocervix

Causative organisms

- *Chlamydia trachomatis* - the most common cause
- *Neisseria gonorrhoeae*



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Risk factors for PID

- Age 25
- Previous STIs
- New sexual partner/multiple sexual partners
- Uterine instrumentation such as surgical termination of pregnancy
- Intrauterine contraceptive devices
- Post-partum endometritis

Features

- lower abdominal pain/ adnexal tenderness
- fever
- deep dyspareunia
- dysuria and menstrual irregularities may occur
- vaginal or cervical discharge
- cervical excitation
- perihepatitis (Fitz-Hugh Curtis Syndrome) occurs in around 10% of cases. It is characterised by right upper quadrant pain and may be confused with cholecystitis

Investigation

- screen for Chlamydia and Gonorrhoea

Management

- Outpatient : oral ofloxacin + oral metronidazole **OM**
or intramuscular ceftriaxone + oral doxycycline + oral metronidazole
- Inpatient: intramuscular ceftriaxone + doxycycline + metronidazole **CDM**
- Removal of any IUD >> better outcome

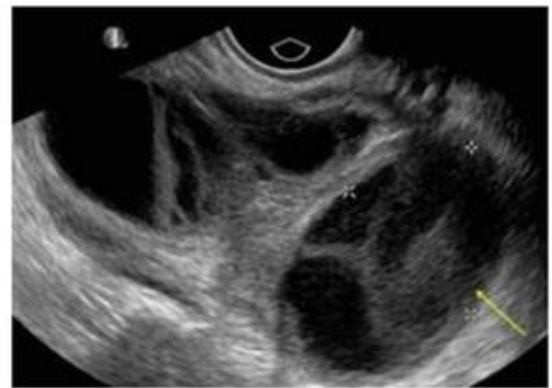
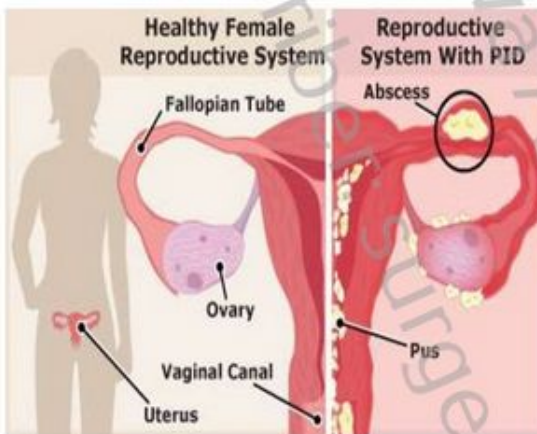


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Complications

- **infertility** - the risk may be as high as 10-20% after a single episode
- **chronic pelvic pain**
- **ectopic pregnancy**
- **tuboovarian abscess** (↑temp, persistent symptoms+ NO DISCHARGE)



Seen in untreated PID or ineffective treatment

If you suspect tuboovarian abscess >>> pelvic US should be done NOT SWABS (take days)

Examoscope:

A female with vaginal discharge, deep seated pelvic pain/ dyspareunia/ +ve cervical motion tenderness >>> diagnosis ?? treatment??

** Same scenario, she started antibiotics 10 days ago, now came with fever, abdominal pain and No discharge >>> investigation?? Diagnosis?



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13- Cervical ectropion:

Normally: On the ectocervix there is a transformation zone where the stratified squamous epithelium meets the columnar epithelium of the cervical canal.

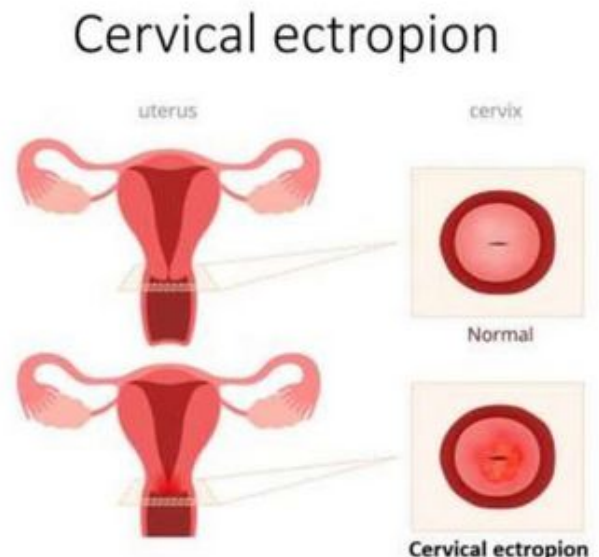
Elevated oestrogen levels (ovulatory phase, pregnancy, combined oral contraceptive pill use) >>> Cervical ectropion (The stratified squamous epithelium that lines the ectocervix is replaced by columnar epithelium, which has migrated from the endocervix)>> **COLUMNAR IS COMING**

Cervical ectropion - red ring around the os

The cervix enlarges under the influence of oestrogen, and as a result, everts the endocervical canal.

features:

- It is usually asymptomatic
- post-coital bleeding
- excessive non-purulent discharge



****Ablative treatment is only used for bothersome symptoms:**
(cautery with silver nitrate, diathermy and cryotherapy)

- ✓ Squamous to columnar
- ✓ Only post coital bleeding
- ✓ Refer to colposcopy only if causes bleeding on touch
- ✓ Leave it alone (No symptoms = No ttt)



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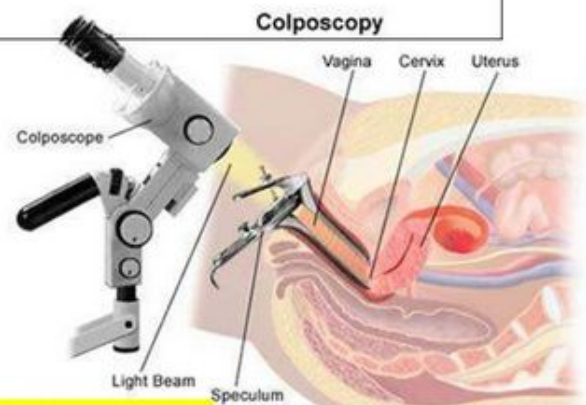
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Caution:

** If this woman is a postmenopausal woman >>> think about endometrial cancer 1st >>> Do transvaginal US

** Don't get tricked by pregnancy >>> if painless and placenta is in normal position >> it is ectopic

indications for a colposcopy:



- ✓ Abnormal cervical cytology on cervical screening (the most common reason for colposcopy referral)
- ✓ Women with symptoms of cervical cancer (e.g. postcoital bleeding or persistent vaginal discharge that infection has been ruled out)
- ✓ Cervical polyps that cause bleeding (note: benign polyps do not need to be removed unless they cause bleeding)
- ✓ Ectropion that causes postcoital bleeding (note: ectropions that do NOT cause bleeding do not need to be referred to colposcopy and requires no treatment)

Examination:

** What is the accurate pathological description for cervical ectropion?

** Female/ young/ on COCP/ postcoital bleeding/ examination >>> friable cx that bleeds on touch >>> next step??

** Female/ young/ post coital bleeding/ normal cervical smear last year/ red cx does not bleed on touch >> management??



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Managing pre-menopausal patients with vaginal bleeding:

According to NICE CKS guidance >>>> if there is abnormal bleeding that persists for more than 6 to 8 weeks, refer to colposcopy even if cervical smears are normal, cervix appears normal and swabs are negative.

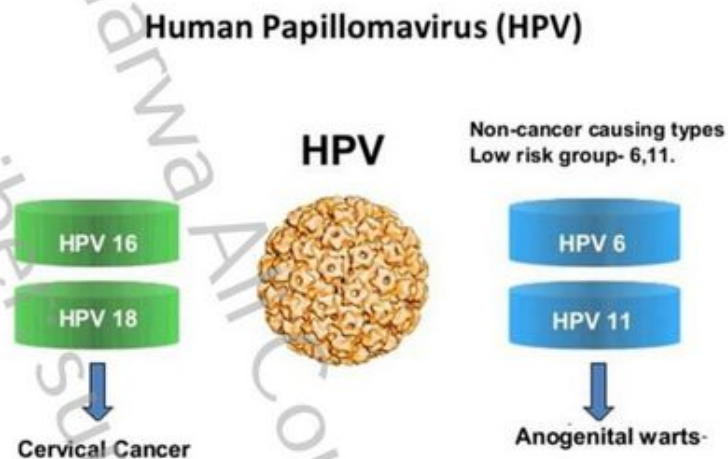
Cervical smears are NEVER used in symptomatic women. Cervical smears are meant for screening asymptomatic women.



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14- Human Papilloma Virus HPV:



Genital warts: Sexually transmitted painless growth- like lesions "benign epithelia skin tumours".

- ✓ HPV 6 and 11 >>>> Responsible for Genital warts (benign Cauliflower like- growths).
- ✓ HPV 16 and 18 >>>> Responsible for most cervical cancers



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Prevention and Treatment of Genital Warts:

>>> Gardasil : for prevention. (A vaccine to protect against HPV 6, 11, 16 and 18). (If genital warts have developed, Gardasil is of no benefit)

>>> 30% of cases >>> spontaneous resolution

>>> Ablation (Cryotherapy) for treatment, other options include podophyllotoxin, imiquimod and excision under local anaesthetic..



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15- Cervical cancer:

Around 50% of cases of cervical cancer occur in women under the age of 45 years, with incidence rates for cervical cancer in the UK are highest in people aged 25-29 years

- squamous cell cancer (80%)
- adenocarcinoma (20%)

Features:

- may be detected during routine cervical cancer screening
- abnormal vaginal bleeding: postcoital, intermenstrual or postmenopausal bleeding / vaginal discharge

Cervical cancer screening program:

The main aim of cervical screening is to detect pre-malignant changes rather than to detect cancer.

A smear test is offered to all women between the ages of 25-64 years

- 25-49 years: 3-yearly screening
- 50-64 years: 5-yearly screening

The optimum time for a cervical smear should be on days 10-20 of a 28-day cycle as most endocervical cells can be collected then. During mid-cycle, there are few polymorphs and more mature endocervical cells. It is also more challenging to obtain endocervical smears when the women are menstruating. For this reason, women are informed to make an appointment during the middle of their menstrual cycle.



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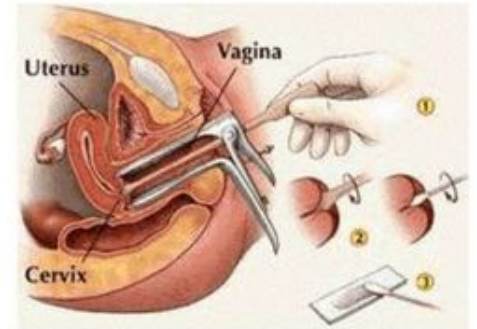
Pap smear-cervical smear : cytology and HPV

Cervical cancer screening: interpretation of results

Cervical intraepithelial neoplasia is abbreviated to CIN

OLD:

Cytology



Result	Management
Inflammatory without dyskaryosis	repeat in 6 months
Borderline or mild dyskaryosis	The original sample is tested for HPV* - if negative the patient goes back to routine recall - if positive the patient is referred for colposcopy
Moderate dyskaryosis	Consistent with CIN II. Refer for urgent colposcopy (within 2 weeks)
Severe dyskaryosis	Consistent with CIN III. Refer for urgent colposcopy (within 2 weeks**)
Suspected invasive cancer	Refer for urgent colposcopy (within 2 weeks)
Inadequate	Repeat smear - if persistent (3 inadequate samples), assessment by colposcopy



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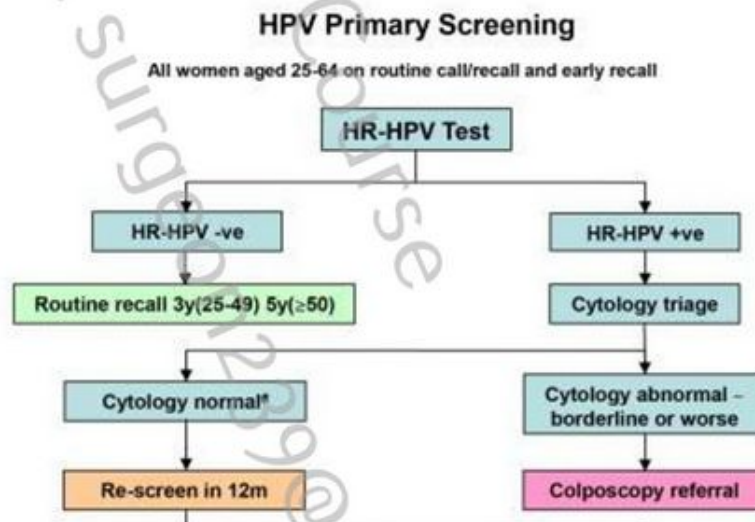
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NEW:

HPV would be the primary screening test in replace of liquid-based cytology.

The reasons for that is primary HR-HPV testing has higher sensitivity for high grade cervical intraepithelial neoplasia (CIN) than primary cytology. HR-HPV testing has a lower rate of false-negative compared to cytology.

This is how it will look like.



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16- Pelvic congestion:

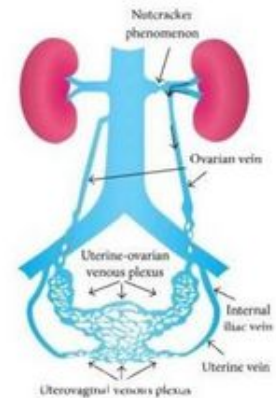
- Dysmenorrhoea
- Dyspareunia
- Increase on Prolonged standing

CLINICAL FINDINGS

- 20-30 year females, multiparous
- Disappear after menopause

➤ Symptoms
chronic pelvic pain, heaviness, dyspareunia, dysmenorrhoea, urinary and rectal symptoms, relieved on supine position.

increase by prolonged standing



Exam scope: a female waitress, C/O of deep dyspareunia/ dysmenorrhoea/ pain worsens on prolonged standing/ laparoscopy is normal >>> diagnosis??

Premenstrual syndrome:

Emotional and physical symptoms that women may experience prior to menstruation.

Common symptoms include:

- anxiety
- stress
- fatigue
- mood swings

Best contraception >>> COCP



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17- Uterine fibroids:

- Benign smooth muscle tumours of the uterus.
- more common in Afro-Caribbean women
- rare before puberty >>> develop in response to oestrogen (Don't tend to progress following menopause)

Features:

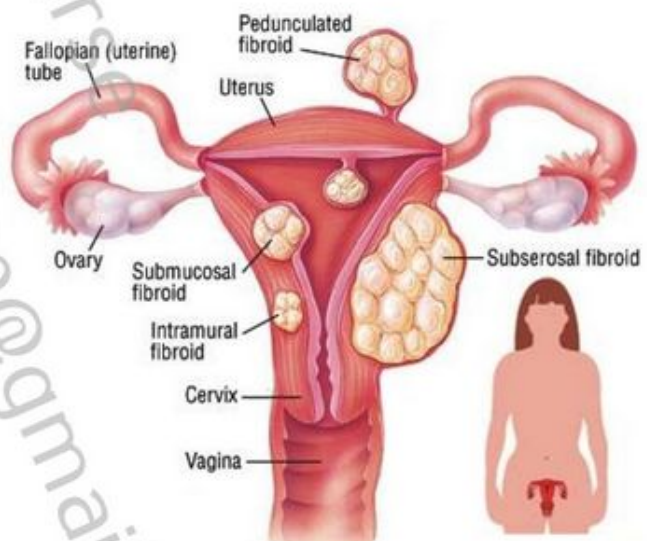
- may be asymptomatic
- menorrhagia
- lower abdominal pain: cramping pains, often during menstruation
- bloating
- urinary symptoms, e.g. frequency, may occur with larger fibroids
- subfertility

Types :

- * submucosal
- * subserosal
- * intramural

Diagnosis:

- TV ultrasound



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Management:

- **Asymptomatic** >>> follow up annually (rapid growth = suspect malignancy and further investigations)
- **Menorrhagia** >>> levonorgestrel-releasing intrauterine system (IUS) is recommended as first-line (in fibroids not distorting uterine cavity)

other options include tranexamic acid, combined oral contraceptive pill, NSAIDs

norethisterone as a temporary measure
- **Severe menorrhagia + large fibroid (>3 cm)** >>> Ulipristal acetate (ella one pills) up to 4 courses (each is 3 m), used for pre operative ttt
- **Surgery** >>> is sometimes needed:
 - ✓ hysterectomy (the best option) >> if completed her family
 - ✓ myomectomy (the best to preserve fertility) >> if want children
 - ✓ uterine artery embolization (preserve fertility too)
 - ✓ endometrial ablation (if <3) but do not preserve fertility,
- **Pre-operative** >>> GnRH agonists :may reduce the size of the fibroid and so decreases the blood loss during operation, but they will return to pre-treatment size, if stopped

Management: (medical/ surgical/ follow up)



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18- Endometrial hyperplasia :

Endometrial hyperplasia is a premalignant condition, that can predispose to, or be associated with, endometrial carcinoma.

- overgrowth of endometrial cells caused by excess unopposed oestrogens, either endogenous or exogenous, similar to endometrial cancer, with which it shares a common aetiology

Presentation : Usually presents clinically as abnormal vaginal bleeding (intermenstrual, polymenorrhoea or postmenopausal)

- It is most commonly diagnosed in women over 40 years old with irregular menstruations or in those with post-menopausal bleeding

Investigation :

- ✓ Transvaginal ultrasound (TVUS) is an appropriate first-line procedure. (the thicker the endometrium seen on ultrasound, the higher the likelihood endometrial cancer (normal: $< 4 \text{ mm}$))
- ✓ Endometrial sampling or formal endometrial curettage is necessary for diagnosis (diagnostic/definitive).

Management of endometrial hyperplasia (no atypia) : depends on age of patient, histology, symptoms, and desire for retaining fertility.

Both continuous oral progestogens and levonorgestrel-releasing IUS are effective in achieving regression of endometrial hyperplasia without atypia. (IUS is preferred as per Royal College).



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19- Endometrial cancer:

Endometrial Cancer

Most common gynecologic malignancy in developed countries

The risk factors:

- obesity
- nulliparity
- early menarche
- late menopause
- diabetes
- PCO
- **unopposed oestrogen.**



The addition of a progestogen to oestrogen reduces this risk (e.g. In HRT). The BNF states that the additional risk is eliminated if a progestogen is given continuously

- **tamoxifen**

Features:

- **postmenopausal bleeding** is the classic symptom
- premenopausal women may have a change intermenstrual bleeding

Investigation:

- **first-line** investigation is **trans-vaginal ultrasound** - a normal endometrial thickness (< 4 mm)
- **definitive** is >>>> hysteroscopy with endometrial **biopsy**



Clinical

- Abnormal uterine bleeding
- Postmenopausal (most common)
- Increasing age in premenopausal

Diagnosis

- Transvaginal ultrasound (postmenopausal women)
- Endometrial sampling



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examoscope:

** A female on tamoxefin/ what is the most alarming sign??

** 60 yrs female + post coital bleeding >>> next step??

** 55 yrs lady/ vaginal bleeding >>> TVUS reveals endometrial thickness 8 mm >>> next step??



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20- Cessation of Menstruation:

Perimenopause vs Menopause

Comparison Chart

Characteristics	Perimenopause	Menopause
Definition	Period of a gradual decline of the estrogen levels in the woman's body prior to the menopause.	Period in the woman's life when menstruation ceases. This is the end of the ability to conceive and give birth.
Age	The beginning of the woman's 40s.	The beginning of the woman's 50s.
Symptoms	Irregular period, fertility issues, worse PMS, heavier or lighter period than normal, weight gain, breast tenderness, increased heartbeat, headache, loss of sex drive, concentration difficulties, forgetfulness, urinary tract infections, muscle pain, hair changes.	Lack of period, hot flashes, night sweats, mood swings, anxiety, irritability, depression, fatigue, insomnia, dry skin, vaginal dryness, frequent urination, incontinence.



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Menopause

Menopause is defined as the permanent cessation of menstruation

Menopause is a clinical diagnosis usually made when a woman has not had a period for 12 months.

The average age in the UK is 51 years old. (usually occur between 45-55)

symptoms

Are caused by ↓ levels of female hormones, principally oestrogen:

- Vasomotor symptoms :hot flushes - night sweats
- Urogenital changes :vaginal dryness and atrophy - urinary frequency
- Psychological: anxiety and depression may be seen - short-term memory impairment
- Longer term complications: osteoporosis - ↑ risk of IHD
- change in length of menstrual cycles
- dysfunctional uterine bleeding may occur

Management:

The management of menopause can be split into three categories:

- Lifestyle modifications
- Hormone replacement therapy (HRT) (uterus/No uterus (cyclic/continuous)
- Non-hormone replacement therapy (e.g SSRIs)



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HRT

Types:

1- Oestrogen-only :

- ✓ Had a **hysterectomy**
- ✓ Have an **IUS**

2- combined HRT

a- Sequential(cyclical combined HRT):

- ✓ Is offered to **peri-menopausal women** is to ↓ irregular bleeding.
- ✓ contains oestrogen which is taken every day and progesterone which is added in the last 12 to 14 days of the menstrual cycle to trigger a bleed.

b- Continuous combined HRT :

- ✓ **Menopausal** women
- ✓ Women who have been **using sequential combined HRT for a year** (as the risk of irregular bleeding is minimal at that point)
- ✓ has the same amount of oestrogen and progesterone throughout the cycle and women should not expect a bleed with it (since they would be postmenopausal when using it).



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Indications of HRT:

- Vasomotor symptoms such as flushing, including sleep, mood disturbance and headaches
- early menopause >>> treated with HRT until the age of natural menopause (around 51 years) - this is to prevent the development of osteoporosis
- For those women <60 years who are at risk of an osteoporotic fracture in whom non-oestrogen treatments are unsuitable

Contraindications:

- Any oestrogen-sensitive cancer
- Current or past breast cancer
- Undiagnosed vaginal bleeding in a post menopausal woman
- Untreated endometrial hyperplasia
- Previous Venous thromboembolism (unless the woman is on anticoagulants)/ **No increased risk with transdermal HRT.**
- Stroke: slightly increased risk with oral oestrogen HRT.
- Active or recent Coronary heart disease
- Untreated hypertension
- Liver disease with abnormal LFT



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Don't get confused between terms:

Key definitions

***Perimenopausal**: Women who are within 12 months of their last period. This is the **transition phase** preceding menopause and it ends at 12 months from their last period. Usually starts in mid-to-**late-40s** and can last for 4 to 5 years.

***Menopausal**: Women who are **12 months or more from their last period**

Do NOT use early menopause and premature ovarian failure interchangeably

early menopause: is used for those women who go through their menopause between **40-45 years**

Do NOT use premature menopause and premature ovarian failure interchangeably

Premature ovarian failure: is sometimes referred to as **premature menopause**, but the two conditions aren't exactly the same.

Women with **premature ovarian failure** may have irregular or occasional periods for years and **may even become pregnant**.

Women with premature menopause: stop having periods and can't become pregnant.



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Atrophic vaginitis:

Atrophic vaginitis often occurs in women who are post-menopausal women. It presents with vaginal dryness, dyspareunia and occasional spotting $\pm$ urinary symptoms

On examination, the vagina may appear pale and dry.

Treatment is with topical oestrogen cream



Atrophic vaginitis and Vulvovaginal atrophy are the commonest cause of post menopausal bleeding. however the most Worrisome is Endometrial Cancer >>>> so the next step is TVUS



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Lichen sclerosus :

a chronic inflammatory dermatosis affects the skin of the anogenital region in women. It presents with genital **itching** which is worse at night and is seen as **white atrophic plaques**. (little girls or middle-aged women).

It is thought to have an autoimmune mechanism in the pathogenesis. Dyspareunia and pain may be seen in cases of erosions or fissures. **Topical steroids** are used as part of the management.

Follow-up is important at 3 & 6 months (long standing, untreated cases show a 4% incidence of malignant change SCC of the vulva



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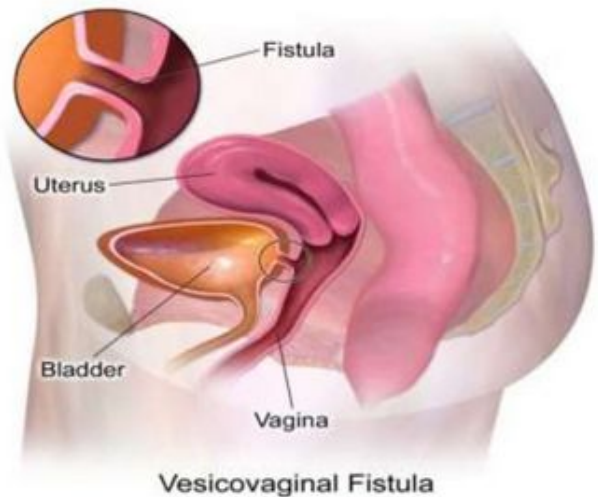
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21- Fistulas:

It is the abnormal connection between urinary and genital tract.

Types: vesicovaginal, urethrovaginal or ureterovaginal

Complaint: continuous leakage of a small amount of clear fluid through the vagina + No distinct odour



RF: after urogenital surgeries e.g hysterectomy

Diagnosis: 3 swab test (methylene blue test)



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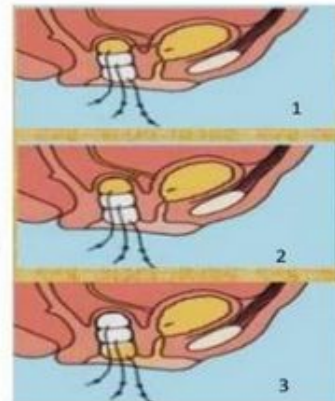
3 swab test (Vaginal Gauze test)

- Three separate sponge swabs are placed into the vagina one above the another
- The bladder is then filled with a coloured agent such as **methylene blue** through a rubber catheter, and patient asked to do some exertional maneuvers and then the swabs are removed after 10 mins.



Result of 3 swab test

1. Discolouration of topmost or middle swab → **vesicovaginal fistula**
2. Uppermost swab wetting but not discolouration → **Ureterovaginal fistula**
3. Discolouration of lower most swab but upper two swabs remain dry → **Urethrovaginal fistula**



Rectovaginal fistula: the passage of foul smelling feculant discharge from the female genital tract (with crohns disease)



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22- Urinary incontinence:

Risk factors

- advancing age (common in elderly)
- multiple pregnancies and vaginal deliveries
- obesity
- family history

Classification

1- Stress incontinence

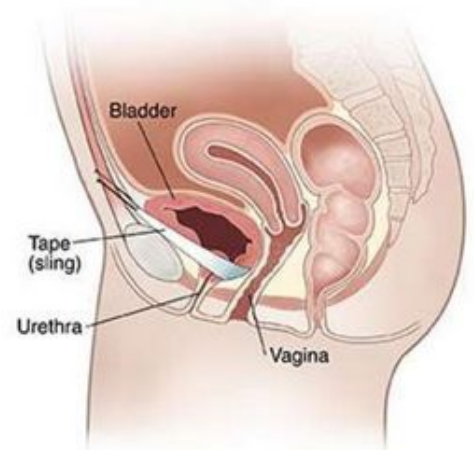
◆ Leakage of urine during **sneezing, coughing, laughing**

◆ cause : **weak bladder outlet** (weak tone) >>> cannot counteract the raised intra-abdominal pressure during activity. Also, with multiple vaginal deliveries >>> Pelvic floor muscles become weak.

◆ Treatment : The initial ttt of choice >>> **Pelvic floor exercise 8X3X3** (at least 8 pelvic contractions, 3 times a day, for at least 3 months).

>>> If failed :Surgical retropubic (mid-urethral tape) (**Free-tension vaginal tape**)

>>>> If surgery is not possible: **Duloxetine**.



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2- Urge incontinence (Detrusor Overactivity)

- ♦ Leakage of urine when there is a sensation of need to void.
(When I feel the desire to pee, I have to go and pee Or: When I feel a desire to pee, sometimes I slightly wet myself "Leakage" before making it to the bathroom)
- ♦ cause : the bladder muscle (**Detrusor**) is **overactive** >>> loss of urine.
- ♦ Treatment : The first line is **behavioural modification** such as modifying fluid intake, avoiding caffeine and alcohol,

Bladder drill (Retraining) gradually increase the periods between voiding (for 6 weeks). >>> re-trains the bladder to void when full or at less frequent intervals >>> better control

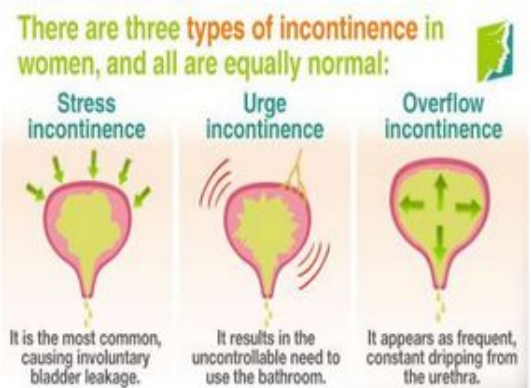
>>>> If drugs are needed : **AntimuscariniC** (e.g. **Oxybutynin**, tolterodine or darifenacin) with continuation of bladder training.

N.B: immediate release oxybutynin should be used with **caution in the elderly** who have increased risk of sudden physical or mental deterioration

3- over flow incontinence:

- ♦ Causes: blockage of outlet, e.g **BPH, urethral stricture**
- ♦ Treatment: **intermittent catheterization**

4- mixed incontinence: both urge and stress



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Contraception:

1- methods of contraception:

- a- Barriers (e.g condoms)
- b- Pills: COCP and POP (short term methods)
- c- Intrauterine devices: IUCD, IUS (Mirena)
- d- Depo-Provera (IM medroxyprogesterone acetate)
- e- Nexplanon (etonogestrel implant)(POSDIS)
- f- Tubal ligation (risk of pregnancy 1:200)
- g- Emergency contraception

2- Choosing a contraceptive method:

- a- Age (above or below 20)
- b- Long term/short term
- c- Post-partum contraception No 6
- d- Presence of co morbidities
 - Fibroids: distorting/not
 - Uterine anomalies
 - Irregular menses
 - Learning difficulties
 - DM, H+, obesity , smoking, migraine with aura
 - PMS
 - Sickle cell disease
 - epilepsy

3- Contraception related problems:

- a- Unscheduled bleeding
- b- Post pill amenorrhea
- c- Lost IUD threads
- d- Failure rate



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• **Methods of contraception**

Contraceptive	Method of action	Notes
Condoms	Physical barrier	✓ Relatively low success rate ✓ Help protects against STIs ✓ ↓ risk of cervical Cx (HPV)
Combined oral contraceptive pill	Inhibits ovulation	✓ ↑ risk of venous thromboembolism and breast cancer
Injectable contraceptive (medroxyprogesterone acetate)	Inhibits ovulation	✓ Lasts 12 weeks
Implantable contraceptive (etonogestrel)	Inhibits ovulation	✓ Last 3 years ✓ Irregular bleeding a common side-effect
Intrauterine contraceptive copper device	Inhibit implantation	Used in migraine with aura
Intrauterine system (levonorgestrel)	Prevents endometrial proliferation	✓ Irregular bleeding a common side-effect



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Failure rates (pearl index)

Method	Typical use	Perfect use (consistently & correctly)
Progesterone-only implant	0.05%	0.05%
Vasectomy	0.15%	0.10%
Mirena (IUS)	0.2%	0.2%
Tubectomy	0.5%	0.5%
Copper IUD	0.8%	0.6%
Injectable Progesterone	6%	0.2%
COCP & POP	9%	0.3%
Male condom	18%	2%



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Combined oral contraceptive pill:

Advantages	Disadvantages
effective and reversible upon stopping (short term)	people may forget to take it offers no protection against STIs
usually makes periods regular, lighter and less painful	↑ risk of venous thromboembolic disease, of stroke and ischaemic heart disease
↓ risk of ovarian, endometrial Cx	↑ risk of breast and cervical cancer

contraindications: (UKMEC) criteria:

- **UKMEC 1:** no restriction for the use ✓✓✓✓
(e.g breastfeeding after 6 months)
- **UKMEC 2:** advantages generally outweigh the disadvantages ✓✓✓X
(e.g between 6 wks and 6 months postpartum in breastfeeding)
- **UKMEC 3:** disadvantages generally outweigh the advantages ✓XXX
- **UKMEC 4:** unacceptable health risk XXXX
(e.g migraine with aura, less breastfeeding < 6 weeks post-partum)



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Contraindications:

- Obesity (BMI > 30) (UKMEC 2)
- Age > 40 (UKMEC 2)
- smoking (Smoker or ex-smoker (UKMEC 2)
- hypertension (even if well controlled)
- history or FHx of thromboembolic disease, stroke or ischaemic heart disease
- migraine with aura (**UKMEC 4**) >>> the only UKMEC 1 is IUCD , the rest are UKMEC 2
- current breast cancer or gene carrier or Hx of breast cancer
- major surgery with prolonged immobilisation
- post partum (non breast feeding for 6 weeks – breast feeding for 6 months)
- learning difficulties

Post pill amenorrhea:

- no resumption of normal menstrual periods after stoppage of OCP.
- **Investigations** are only required **after 6 months** have passed from stopping oral contraception where the woman still has no periods.

Laboratory findings: NORMAL FSH/LH/Oestrogen, thyroid function
+ May have slightly elevated prolactin

Management :

- Reassure
- Clomiphene >>> if wanting to get pregnant (induce ovulation)

N.B: vaginal spotting is common after initiating hormonal contraception
(e.g COCP >>>> within 3 months >>> reassure
>>>> more than 3 months >>> consider shifting to another method



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Interaction: Enzyme inducers VS inhibitors:

Inducers

(Weakens the effect of pills)

inhibitor

(No effect on pills)

CPP:
carbamazepine/
phenytoin/
phenobarbital

chronic

anti epileptic

sodium valproate

Alcohol

acute

Rifampicin

Anti TB

Isoniazide

grisofulvin

Anti fungal

conazoles
(keto/fluo)

Sulphonylurea

Anti DM

Antibiotics

chloramphenicol
macrolids (eryth/clarith)
sulonamide
ciproflox
Metronidazole

**Anti peptic
ulcer**

cimetidine
omeprazole



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Intrauterine contraceptive devices:

conventional copper intrauterine devices (IUDs)	levonorgestrel-releasing intrauterine systems (IUS, Mirena).
prevention of fertilisation & implantation	+ prevents endometrial proliferation
effective immediately following insertion	after 7 days

Both the IUD and IUS are more than 99% effective

Potential problems

1- IUDs make periods heavier, longer and more painful

The IUS is associated with unscheduled uterine bleeding and spotting.

>>>> reassure, as long as it is not problematic

>>> If problematic>>> try tranexamic acid for 5 days or CoCp for 3 months

>>>> Later >>> intermittent light menses and some women become amenorrhoeic

2- infection: ↑ risk of pelvic inflammatory disease

3- Ectopic pregnancy: (the relative risk of ectopic is increased but the absolute risk is reduced, compared to a woman not using contraception (i.e relative risk ↑, while absolute risk ↓)



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4- Management of lost intrauterine device threads:

IUD thread is not found >>> Speculum examination reveals no IUD thread

- Take menstrual and sexual history, exclude pregnancy.
- Provide alternative contraception and/or post coital contraception if indicated by history
- If women is pregnant >>> Refer to Early Pregnancy Unit for rapid access to ultrasound scan. Viability of pregnancy, site, gestation will need to be determined.
- If women is not pregnant o Refer for ultrasound scan. Ensure contraception is given until ultrasound scan is performed
 - a- If ultrasound scan reports misplaced in cavity, the management depends on symptoms and degree of displacement
 - b- If ultrasound scan reports device is correctly located in the uterus, leave the IUD in situ until it is due to be removed
 - c- If IUD not seen in uterus, order a plain abdominal X-ray
 - >>> If abdo X-ray shows IUD is in the abdominal cavity, this may require laparoscopic removal.
 - >>> If abdo X-ray shows IUD not located, this implies that IUD has fallen out and replacement of IUD can be offered



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Implantable contraceptives

- ✓ inserted in the proximal non-dominant arm, just overlying the tricep.
- ✓ The main mechanism of action is preventing ovulation.

Key points

- long-acting: lasts 3 years
- doesn't contain oestrogen (so can be used if past history of thromboembolism, migraine etc)
- can be inserted immediately following a termination of pregnancy



Injectable contraceptives

Depo Provera is the main injectable contraceptive used in the UK. It contains medroxyprogesterone acetate. It is given via IM injection every 12 weeks.

- ✓ The main method of action is by inhibiting ovulation.
- ✓ delayed return to fertility (maybe up to 12 months)

Adverse effects

**may potentially increased risk of osteoporosis: should not be used < 18 years

** used with sickle cell vvvv
(depo, pop, IUS are UKMEC 1)



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Methods of emergency contraception:

Contraceptive	Mode of action
Levonorgestrel	- Inhibits ovulation - within 72 hours of unprotected sexual intercourse (UPSI)*
Ulipristal (ella one pill)	- Inhibits ovulation or delay ovulation 120 hours after intercourse
Intrauterine contraceptive device	Inhibits implantation and fertilization must be inserted within 5 days of UPSI



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Choosing a contraceptive method:

a- AGE:

Contraceptive Choices for Young People

Legal and ethical issues

- The age of consent for sexual activity in the UK is **16 years**. Practitioners may however provide advice and contraception if they feel that the young person is 'competent'.
- children **under the age of 13 years** are considered unable to consent for sexual intercourse so, consultations regarding this age group >>> child protection measures.
- DO NOT prescribe IUS **below 20 years** old and Depo-Provera **below 18 years** (↑ risk of osteoporosis)
- some concerns about the effect of Depo-provera on **bone mineral density** under the age of 18 years and the IUS and IUD for women under the age of 20 years (UKMEC 2, meaning they may not be the best choice)
- COCP and POP are safe below 20 years
- Nexplanon are safe below 20 years

Patient > 20 >>> XXX IUS/IUCD (UKMEC 2) >> (jaydess)

Patient > 18 >>> XXX depo-provera (UKMEC 2)

b- long / short term:

- ✓ long-acting reversible contraceptive methods : IUS and progesterone only implants
- ✓ short term methods: pills, barriers



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c- Post-partum contraception

After giving birth women require contraception **after day 21**. (First 21 days >>> No contraception)

Lactational amenorrhea criteria:

- ✓ Less than 6 m from the birth of the baby
- ✓ Amenorrheic
- ✓ FULLY Breast feeding

➤ Progestogen only pill (POP)

- breastfeeding and non-breastfeeding >>> can start the POP at any time postpartum.'

➤ Combined oral contraceptive pill (COC)

- may be started from day 21 >> provide immediate contraception
- **absolutely contraindicated - UKMEC 4** - if breast feeding < 6 weeks post-partum (↑risk of thromboembolism)
- **UKMEC 2 - if breast feeding 6 weeks - 6 months** postpartum (↓ milk production)
- After 6 wks + NOT breastfeeding >>> COCP

➤ IUS:

Can be used within 48 hours of delivery or after 4 wks (for fear of perforation) useful only if long term contraception is needed

examoscope: A 30 year old lady / 4 weeks post partum / breast feeding. She wishes to re-start contraception / would like to have more children in the near future/ she is needle phobic. What is the most appropriate contraception



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d- Presence of co morbidities

Fibroids: distorting/not >>>> ???

- ✓ Fibroids distorting uterine cavity >>> XXX IUS (UKMEC 3)
- ✓ Fibroids NOT distorting uterine cavity >>> IUS >> shrink their size

Uterine anomalies >>>> ???

Irregular menses >>>> ???

- ✓ Menorrhagia >> IUS or COCP
- ✓ Menorrhagia >> XXX IUCD / implanon

Learning difficulties >>>> Nexoplanon

DM, H+, obesity, smoking, migraine with aura >>>> ???

PMS >>>> COCP

History of VTE >>> IUCD (UKMEC 1) / IUS (UKMEC 2)

Epilepsy and contraception:

Take care of: the effect of the anti-epileptic drugs on the effectiveness of the contraceptive (**enzyme inducers**)

Take care of: the **teratogenic** effects of the anti-epileptic if the woman becomes pregnant >>>> use condoms+ other forms of contraception for women taking anti epileptic drugs

With Na Valproate >> all are UKMEC 1 (DON'T USE CONDOMS)

Sickle Cell:

- ✓ UKMEC 1 >>> **depoprovera** / POP / IUS/ progesterone implants
- ✓ UKMEC 2 >>> COCP/ IUCD



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Menstrual irregularities / doesn't require contraception:

Menorrhagia management

Does not require contraception

Menorrhagia only >>>> tranexamic acid 1 g tds

Menorrhagia + Dysmenorrhea >>>> Tranexamic+ NSAIDs e.g
Mefenamic acid

Metrorrhagia (irregular) +/- >>>>> COCP

Requires contraception:

- 1st line >>> intrauterine system (Mirena)
- combined oral contraceptive pill
- long-acting progestogens

Norethisterone 5 mg tds can be used as a short-term option to rapidly stop heavy menstrual bleeding.



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